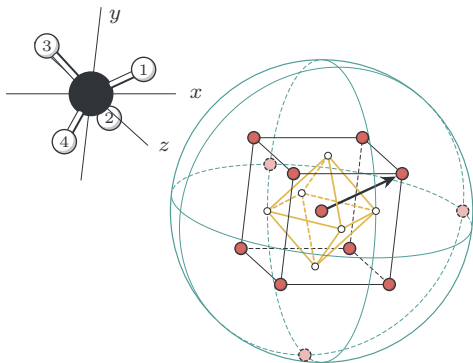
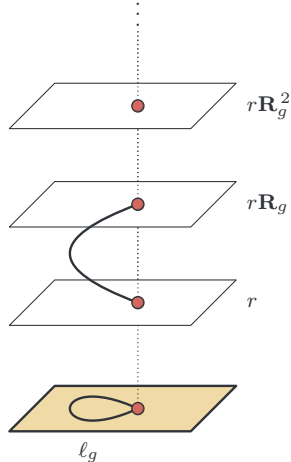


methane, CH_4 : the reference $\mathbf{X}_0$ in its body frame,
the C_2 axes x, y, z through the midpoints of opposite edges



the orientation space $SO(3)$ as a ball,
direction the axis and distance the angle:
the twelve rotations of T in red,
the cell $\mathcal{U}$ at $q=0$ in gold,
and the lift $r \rightarrow r\mathbf{R}_g$ of the
third-turn about bond 1



over the base $SO(3)/T$, twelve sheets,
one for each rotation in T , three drawn;
the third-turn closes as a loop ℓ_g in the base
and, lifted, climbs from sheet r to sheet $r\mathbf{R}_g$;
A sees the climb as 1, ${}^1\text{E}$ as ω ,
 ${}^2\text{E}$ as ω^2 , T as $x \rightarrow y \rightarrow z$